

Name:

ID #

McGill University

STATISTICS (MATH324)

Mid-Term Exam, Tuesday October 22nd, 2019

Time: 11:30 - 13:00

Instructions: This test consists of four questions. There is no optional question. Please answer all four questions. A two-sided 8.5×11 page of formula and a calculator are permitted. The following information may be of use to you:

- $X \sim \text{Normal}(\mu, \sigma^2) :$ $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ $x \in \mathbb{R}$
- $X \sim \text{Poisson}(\lambda) :$ $p_X(x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x \in \{0\} \cup \mathbb{N}$
- $X \sim \text{Binomial}(n, p) :$ $p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$
- $\text{Bernoulli}(p) = \text{Binomial}(1, p)$
- For $Z \sim N(0, 1)$, $P(|Z| < 1.96) = 0.95$.

Good luck.

Question 1. In polycrystalline aluminum, the number of grain nucleation sites per unit volume is modelled as having a Poisson distribution with mean λ . Fifty unit-volume test specimens subjected to annealing under regime A produced an average of 20 sites per unit volume. Fifty independently selected unit-volume test specimens subjected to annealing regime B produced an average of 23 sites per unit volume.

(a) Estimate the difference in the mean numbers of nucleation sites $\lambda_A - \lambda_B$ for regimes A and B, and find a 95% confidence interval for it. **(5 marks)**

Solution:

Let $X_i \sim Po(\lambda_A)$, $i = 1, 2, \dots, m$ and $Y_j \sim Po(\lambda_B)$, $j = 1, 2, \dots, n$, where $m = n = 50$. Since λ_A and λ_B are respectively the expected values of X 's and Y 's a 95% confidence interval for $\lambda_A - \lambda_B$ is

$$\bar{X}_m - \bar{Y}_n \pm z_{\alpha/2} \sqrt{\frac{\sigma_A^2}{m} + \frac{\sigma_B^2}{n}} \quad , \quad (1)$$

where $\alpha = 1 - 0.95 = 0.05$, $\sigma_A^2 = \lambda_A$ and $\sigma_B^2 = \lambda_B$, while $\bar{X}_m$ and $\bar{Y}_n$ are respectively the sample averages of X 's and Y 's. The sample averages, $\bar{X}_m$ and $\bar{Y}_n$ are therefore consistent estimator of σ_A^2 and σ_B^2 respectively. Having replaced σ_A^2 and σ_B^2 respectively by $\bar{X}_m$ and $\bar{Y}_n$ in (1) we have

$$\bar{X}_m - \bar{Y}_n \pm z_{\alpha/2} \sqrt{\frac{\bar{X}_m}{m} + \frac{\bar{Y}_n}{n}} \quad .$$

Now plugging-in for $\bar{X}_m = 20$, $\bar{Y}_n = 23$, $m = n = 50$ and $z_{\alpha/2} = 1.96$ we obtain

$$20 - 23 \pm 1.96 \sqrt{\frac{20}{50} + \frac{23}{50}} \approx -3 \pm 1.82 = (-4.82, -1.18).$$

Since $0 \notin (-4.82, -1.18)$, the above confidence interval indicates that λ_A is most likely not equal λ_B and, in fact, $\lambda_A < \lambda_B$.

(b) If the estimate of the difference in the mean numbers is to be correct to within one site per unit volume with probability 0.95, how many specimens must be annealed under each regime? **(5 marks)**

Solution:

To determine the required sample size we need to solve the following equation

$$z_{\alpha/2} \sqrt{\frac{\sigma_A^2}{m} + \frac{\sigma_B^2}{n}} = B \quad .$$

It is clear that with two unknowns, m and n , and only one equation we can have infinitely many solutions. In real applications we often choose m and

n in such a way that $m/n = k$ where k is a known value. A common choice for k is 1. Having solved the above equation we have

$$n = \left(\frac{z_{\alpha/2}}{B}\right)^2 \frac{\sigma_A^2 + k\sigma_B^2}{k} \quad .$$

Having assumed that $k = 1$ we obtain

$$m = n = \left(\frac{z_{\alpha/2}}{B}\right)^2 (\sigma_A^2 + \sigma_B^2) \quad .$$

Plugging-in for $z_{\alpha/2} = 1.96$, $B = 1$, $\sigma_A^2 = 20$ and $\sigma_B^2 = 23$, we have $m = n = 165.19$. We need therefore have $m = n = 166$ specimens.

Question 2. Suppose that $Y_1, \dots, Y_n$ is a random sample from a probability density function in the (one-parameter) exponential family so that

$$f(y; \theta) = \begin{cases} a(\theta)b(y)e^{-c(\theta)d(y)}, & \text{if } \alpha \leq y \leq \beta \\ 0, & \text{otherwise,} \end{cases}$$

where α and β do not depend on θ .

(a) Show that $\sum_{i=1}^n d(Y_i)$ is sufficient for θ . (5 marks)

Solution:

We need first find the likelihood of the observations:

$$\mathcal{L}(\theta; y_1, \dots, y_n) = \prod_{i=1}^n a(\theta)b(y_i)e^{-c(\theta)d(y_i)} = \left\{ \left(a^n(\theta)e^{-c(\theta)\sum_{i=1}^n d(y_i)} \right) \left(\prod_{i=1}^n b(y_i) \right) \right\}.$$

Using Fisher-Neymann factorization, $\sum_{i=1}^n d(y_i)$ is a sufficient statistic for θ .

(b) Show that Binomial (n, p) is a member of the exponential family. Using part (a) find a sufficient statistics for p . (5 marks)

Solution:

If $Y \sim \text{Bin}(n, p)$, then

$$P(Y = y) = \binom{n}{y} p^y (1-p)^{n-y} = (1-p)^n \binom{n}{y} e^{[\log(\frac{p}{1-p})]y} \quad \text{for } 0 \leq y \leq n, y \in \mathbb{N}.$$

Comparing the above form of Binomial distribution with the form given in part (a), we have $\theta = p$, $a(\theta) = (1-p)^n$, $b(y) = \binom{n}{y}$, $c(\theta) = -\log(\frac{p}{1-p})$ and $d(y) = y$. Thus $\sum_{i=1}^n y_i$ is a sufficient statistic for p in Binomial model.

Question 3. Suppose that $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_n$ are independent random samples from normal distributions with means μ_1 and μ_2 and common variance σ^2 .

(a) Show that $\hat{\sigma}^2 = \left\{ \sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (Y_i - \bar{Y})^2 \right\} / (2n - 2)$ is a consistent estimator of σ^2 . (5 marks)

Solution:

First notice that

$$\hat{\sigma}^2 = \frac{1}{2} [S_X^2 + S_Y^2] \text{ where } S_X^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \text{ and } S_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

We learnt in class that

$$S_X^2 = \left(1 + \frac{1}{n-1}\right) \left[\frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \right] \rightarrow 1 \times [\mathbb{E}(X^2) - \{\mathbb{E}(X)\}^2] = \text{Var}(X).$$

The above result follows from Kolmogorov's theorem and Theorem 9.2(a, b and d). Given that Xs and Ys have a common variance σ^2 , S_Y^2 also converges in probability to σ^2 . Then using Theorem 9.2(a), $S_X^2 + S_Y^2 \rightarrow 2\sigma^2$ and using Theorem 9.2(b), $\hat{\sigma}^2 \rightarrow \sigma^2$. Note that this proof does not require Normality assumption. So as long as Xs and Ys are iid with a common variance σ^2 , $\hat{\sigma}^2$ is a consistent estimator of σ^2 .

(b) Is $\hat{\sigma}^2$ given in part (a) a UMVUE? Why? (5 marks)

Solution:

First note that given what we showed in class S_X^2 is an unbiased estimator of σ^2 and so is S_Y^2 since Xs and Ys have common variance. Thus

$$\mathbb{E}(\hat{\sigma}^2) = \frac{1}{2} [\mathbb{E}(S_X^2) + \mathbb{E}(S_Y^2)] = \frac{1}{2} [\sigma^2 + \sigma^2] = \sigma^2.$$

On the other hand, the likelihood is

$$\begin{aligned} \mathcal{L}(\mu_1, \mu_2, \sigma^2; x_1, \dots, x_n, y_1, \dots, y_n) &= \prod_{i=1}^n \frac{e^{-\frac{(x_i - \mu_1)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}} \prod_{i=1}^n \frac{e^{-\frac{(y_i - \mu_2)^2}{2\sigma^2}}}{\sigma\sqrt{2\pi}} \\ &= (2\pi\sigma^2)^{-n} \exp \left\{ -\frac{1}{2\sigma^2} \left[\sum_{i=1}^n (x_i^2 + y_i^2) - 2\mu_1 \sum_{i=1}^n x_i - 2\mu_2 \sum_{i=1}^n y_i + n(\mu_1^2 + \mu_2^2) \right] \right\}. \end{aligned}$$

Thus using Fisher-Neyman Theorem $(\sum_{i=1}^n X_i, \sum_{i=1}^n Y_i, \sum_{i=1}^n (X_i + Y_i))$ is a sufficient statistic for (μ_1, μ_2, σ^2) . It is not hard to see that there is a one-to-one correspondence between $(\sum_{i=1}^n X_i, \sum_{i=1}^n Y_i, \sum_{i=1}^n (X_i + Y_i))$ and $(\bar{X}, \bar{Y}, \hat{\sigma}^2)$ implying that the latter vector is also sufficient for (μ_1, μ_2, σ^2) . Now $\hat{\sigma}^2$ is clearly a function of the sufficient statistic. Given that it is also unbiased, it is the UMVUE. Note that the sufficient statistic in this case is also complete. Completeness is an important requirement which I talked about in class.

Question 4. Suppose $X_i \stackrel{iid}{\sim} \text{Poisson}(\lambda)$, for $i = 1, 2, \dots, n$. Find the maximum likelihood estimator (MLE) of $P_\lambda(X = 0)$. **(10 marks)**

Solution:

The likelihood of the data is

$$\mathcal{L}(\lambda; x_1, \dots, x_n) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

Taking logarithm of the likelihood, we have

$$\ell(\lambda; x_1, \dots, x_n) = \log \mathcal{L}(\lambda; x_1, \dots, x_n) = -n\lambda + \left(\sum_{i=1}^n x_i \right) \log(\lambda) - \log \left(\prod_{i=1}^n x_i! \right).$$

Taking derivative with respect to λ we find

$$\frac{\partial \ell}{\partial \lambda} = -n + \frac{\sum_{i=1}^n x_i}{\lambda}.$$

Finding the root of the partial derivative we have

$$\hat{\lambda}_{ML} = \bar{X}_n.$$

On the other hand, $P_\lambda(X = 0) = e^{-\lambda}$. Thus using the invariance property of the maximum likelihood approach, the MLE of $P_\lambda(X = 0)$ is $\exp\{-\bar{X}_n\}$.